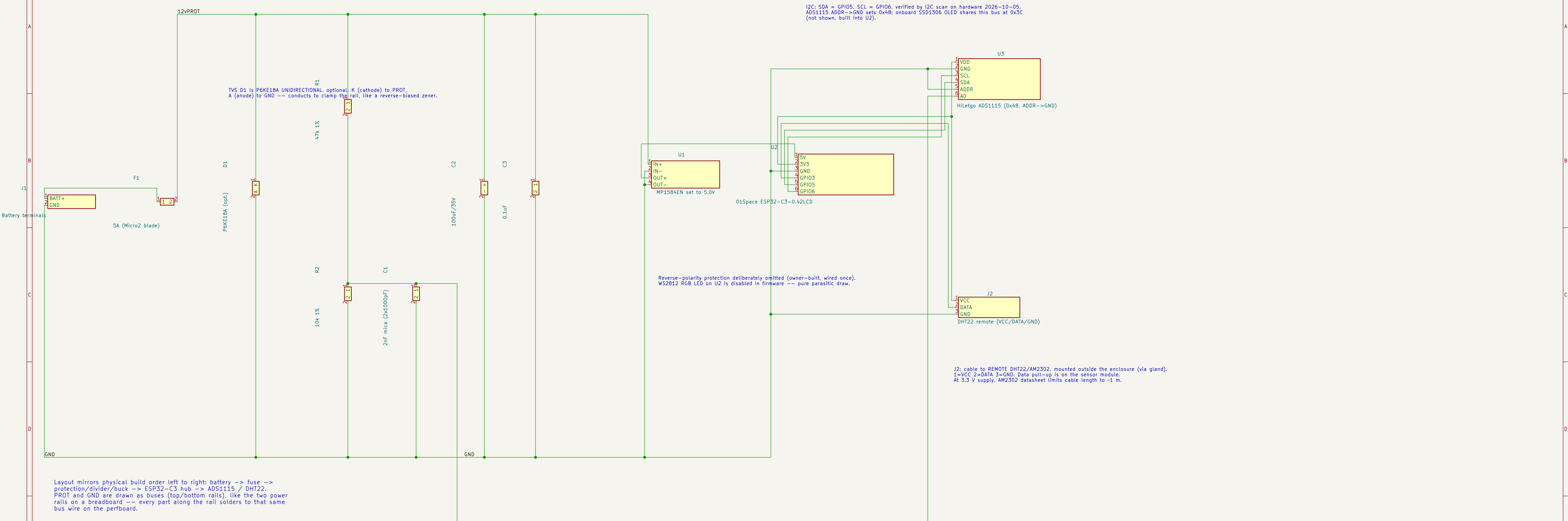


EV 12V BATTERY MONITOR
Battery -> WiFi -> HomeAssistant



Divider R1 47k / R2 10M = 5.71 (unit 1 measured 46.5k/9.85k = 5.72:1). C1 = 2 x 1000pF mica in parallel (2nF, low leakage). ADS1115 abs max = VDD+0.3 = 3.6V.
12.22V(Flooded 50% SOC)-->2.14V 14.4V(charging)-->2.52V 16.0V(design max)-->2.80V @ A0.
ANALOG GROUND: R2 and C1 return to the ADS1115 GND pin, and that node meets power GND at ONE point. Load current through a shared ground path adds directly to A0 and is scaled up 5.7x (bench: 15mV rail-to-rail = -86mV error at the battery).

See CLAUDE.md / PARTS.md in project root for full design rationale
ESP32-C3 + ADS1115 12V battery monitor for 2027 Subaru Uncharted GT
Owner-built

Sheet: /
File: uncharted-battery-monitor.kicad_sch

Title: Uncharted 12V Battery Monitor

Size: A2	Date: 2026-08-29	Rev: B
KiCad E.D.A. 10.0.5		Id: 1/1

Rev: B